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TVPNet: Text-vision prompt guided segmentation for small 3D medical object

Yuhua Xie^a, Rui Wang^b, Yan Zhuang^a, Ke Chen^a, Lin Han^a, Guoliang Liao^a, Yao Hou^a, Linxuan Hou^a, Jiangli Lin^a

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Abstract

Accurate segmentation of small organs in 3D CT images remains a significant challenge due to their small volume, indistinct boundaries, susceptibility to artifacts, and semantic similarity to surrounding tissues. In this study, a text-vision prompt-guided model (TVPNet) is proposed for small target segmentation in 3D medical images, which is built on the SAM framework, integrating a CNN-Transformer image encoder for efficient feature extraction. A pseudo-label decoder structure is also introduced to enhance the model's robustness in segmentation under various prompt input conditions. For small organ segmentation, visual prompts are used to provide explicit region information related to targets, effectively minimizing background interference, while text prompts are employed to decouple the semantics of different organs, improving the model's attention on small targets. Additionally, a multi-combination optimization strategy is proposed during training to enhance the model's understanding of prompt information. By evaluating on the FLARE2023 and MSD datasets, TVPNet achieved an average Dice score of 87.10 on FLARE2023, with a score 82.66 for the segmentation of 8 small organs, surpassing the state-of-the-art baselines for small object segmentation.

Introduction

Medical image segmentation plays a crucial role in clinical applications such as disease diagnosis [1], surgical planning [2], and tumor detection [3], helping to identify tissues, organs or lesions. With the rapid advancement of artificial intelligence, automatic segmentation algorithms based on deep learning have become the dominant method in this field [4], [5], [6], [7], [8]]. Despite their success in various segmentation tasks, these methods still struggle with accurately segmenting small targets, such as tumors and small organs, which are critical for clinical diagnosis.

This challenge is primarily due to the severe class imbalance between small targets and larger structures or background regions in the images. For example, small organs account for less than 0.6% of the CT volume as shown in Fig. 1. Additionally, weak boundaries and background noise further complicate the segmentation of small organs.

The common solution is to use the coarse-to-fine two-stage segmentation framework [4], in which the segmentation process was divided into two stages: the coarse segmentation stage, where the model located the targets and crop the region of interest (ROI), and the fine segmentation stage, where the target was further segmented within the ROI using more refined image features. While this approach enhances segmentation performance for small targets, it comes at the cost of losing important contextual information from the original image, creating a trade-off between focusing on small target details and maintaining the broader image context.

Recently, prompt-based general segmentation models have shown strong interactive segmentation capabilities [5]. Users can guide the segmentation process with textual prompts or spatial prompts, such as points, boxes, and masks. As noted earlier, the two-stage segmentation method relies on prior information obtained from the coarse segmentation stage, which the fine segmentation network then used to improve performance. In contrast, we believe that introducing prior information about small targets through prompts a more direct and efficient solution. Moreover, the variety of prompt types enables the network to capture richer and more relevant spatial semantic information, thereby enhancing segmentation accuracy for small targets.

In this paper, a text-vision prompt-guided model (TVPNet) is proposed for small target segmentation in 3D medical images. TVPNet is designed upon the SAM framework, utilizing textual and visual prompts to enhance the model's attention to small targets. A CNN-Transformer image encoder is designed for effective image feature extraction, while a pseudo-label decoder is proposed to improve the network's robustness by

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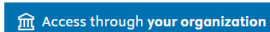
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A pseudo-label decoder is proposed to improve the network's robustness by incorporating prompts generated by pseudo labels during training. In summary, our contributions are as follows:

- (1) We propose a novel prompt-based segmentation framework called TVPNet, which incorporates the advantages of prompt engineering to effectively enhance the network's focus on small targets in 3D medical images. The overall workflow diagram of the framework is shown in Fig. 2.
- (2) A hybrid image encoder and pseudo-label decoder are designed to effectively extract both coarse and fine-grained image features, improving the model's robustness to ambiguous prompts, and enabling effective segmentation even with small-scale datasets.
- (3) Experimental evaluations on the FLARE2023 and MSD challenge datasets demonstrate that TVPNet outperforms state-of-the-art models in small organ segmentation.

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Section snippets

Regular medical segmentation frameworks

Convolutional neural network (CNNs) and Transformer are two commonly used architecture in medical image segmentation. CNN excel at capturing local structural and texture features in images with their inherent locality and translational invariance, which makes it effective for many segmentation tasks and has led to significant advancements in medical image analysis [[6], [7], [8]]. However, these properties also limit CNN when processing global contextual information, which is important for ...

Methods



Fig. 3 illustrates the overall structure of our proposed segmentation network, TVPNet, which adopts an encoder-decoder architecture for segmenting small targets in 3D CT medical images. Specifically, 3D image patches are input into a hybrid CNN-Transformer encoder for feature extraction. Then the extracted image features, along with the spatial prompt embeddings from the prompt encoder and the text prompt embeddings from the CLIP text encoder, are fed into the mask decoder to predict the ...

Dataset and Implementation details

To test the performance and generalization of TVPNet, two public segmentation datasets were used: the FLARE2023 multi-organ segmentation dataset and the pancreatic and colon cancer segmentation dataset from the MSD challenge. The FLARE2023 dataset is a partially labeled dataset, for ease of analysis, 50 3D CT scans (ranging from $0.66 \times 0.66 \times 1.0$ to $0.98 \times 0.98 \times 2.5 \text{ mm}^3$) containing 13 organ labels were selected for experimentation. Based on the volume distribution of various organs in the ...

Conclusion

In this article, we propose TVPNet, a prompt-based segmentation model designed to address the challenge of small organ segmentation. The model leverages a hybrid image encoder for efficient feature extraction from medical images and incorporates prompt engineering techniques to guide the network's focus on small organ targets. By providing prompts, the network acquires prior knowledge about these targets, improving its segmentation accuracy. Experimental results demonstrate that TVPNet ...

CRediT authorship contribution statement

Yuhua Xie: Visualization, Methodology, Conceptualization, Writing – original draft. **Rui Wang:** Data curation, Validation. **Yan Zhuang:** Writing – review & editing, Funding acquisition. **Ke Chen:** Supervision, Investigation. **Lin Han:** Software. **Guoliang Liao:** Investigation, Formal analysis. **Yao Hou:** Formal analysis. **Linxuan Hou:** Validation. **Jiangli Lin:** Writing – review & editing, Supervision. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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